

Network Basics – Day 2

Network

A network is a group of connected devices or systems that can communicate and share resources with each other.

IP Address

An IP (Internet Protocol) address is a unique identifier assigned to a device connected to a network.

IPv4 and IPv6

IPv4 uses 32-bit addresses while IPv6 uses 128-bit addresses, allowing a much larger number of devices to connect to the internet.

Importance of IP Addresses

- Send and receive data
- Access websites
- Connect to the internet
- Identify devices on a network

Types of IP Addresses

Type	Description
Public IP	Provided by the internet service provider and visible on the internet.
Private IP	Used inside home or office networks and assigned by the router.

DNS (Domain Name System)

DNS translates human-readable website names into IP addresses so computers can communicate over the internet.

Record Type	Purpose
A	Maps domain names to IPv4 addresses
AAAA	Maps domain names to IPv6 addresses
MX	Mail server records
CNAME	Alias for another domain
TXT	Verification and security information

Ports

A port is a communication endpoint used by applications and services to exchange data over a network.

Port	Service
20/21	FTP
22	SSH
25	SMTP (Email)
53	DNS
80	HTTP
110	POP3
143	IMAP
443	HTTPS
3389	RDP

Types of Ports

Protocol	Usage
TCP	Reliable communication used for web browsing, email, and file transfer.
UDP	Faster communication used for gaming, streaming, and voice calls.

Port Ranges

Range	Purpose
0–1023	Well-known ports
1024–49151	Registered ports
49152–65535	Dynamic/Private ports

Traceroute

Traceroute is a networking diagnostic tool used to track the path packets take from a device to a destination server. It helps identify delays, routing problems, and connectivity issues.

- Network troubleshooting
- Detecting routing issues
- Finding delays and latency
- SOC investigations
- Connectivity testing

Task: Find the IP Address of 3 Websites

Use commands like **ping**, **nslookup**, to find the IP addresses of websites.

```
winterfell@winterfell:~$ nslookup kseb.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   kseb.com
Address: 91.195.241.232

winterfell@winterfell:~$ nslookup whatsapp.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   whatsapp.com
Address: 57.144.55.32
Name:   whatsapp.com
Address: 2a03:2880:f334:120:face:b00c:0:167

winterfell@winterfell:~$ nslookup chatgpt.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   chatgpt.com
Address: 172.64.155.209
Name:   chatgpt.com
Address: 104.18.32.47
Name:   chatgpt.com
```